

Multi-Branch 30-Day Career Acceleration Roadmap

Module 11: Department-Specific Placement Curriculums

This document compiles the placement readiness and industry-readiness programs for all 10 engineering departments in the PinIT Career OS (V2.0).

1. Computer Science and Engineering (CSE)

Goal: SDE, Full Stack, and Cloud Engineering placement readiness.

1.1 Weekly Curriculum

- Week 1: Assessment & Core Foundations : Career Audit → Resume Engineering (Quantified achievements) → LinkedIn & GitHub optimization → DSA Arrays & Strings (15 pattern problems) → SQL (Joins, Group By, Subqueries) → Assessment.
- Week 2: Core Computer Science : OS (Threads, Scheduling, Deadlocks) → DBMS (Normalization, Transactions, ACID) → Networks (TCP/IP, HTTP, DNS, Load Balancing) → OOP & Design Principles (SOLID) → DSA Linked Lists & Trees.
- Week 3: Industry-Level Development : Project Selection (SaaS, AI App, LMS) → System Design Basics (Scalability, Caching) → Backend API & Auth → DB Design → Cloud Deployment (Docker, CI/CD).
- Week 4: Placement & Growth : Aptitude & Reasoning → Behavioral (STAR framework) → DSA Graphs & DP → Technical/Behavioral Mocks → Open Source → Portfolio.

1.2 Target Companies & Deliverables

- Target Companies : Amazon, Flipkart, Microsoft, Swiggy, Zomato, Razorpay.
- Deliverables : Deployed fullstack app, GitHub portfolio, ATS Resume, 250+ DSA solved.

2. Information Technology (IT)

Goal: Enterprise Software Development, Database Engineering, and Cloud DevOps.

2.1 Weekly Curriculum

- Week 1: Web Tech & SQL : Web Architectures → HTML5/CSS3/ES6 JavaScript → DOM Manipulation & APIs → Advanced SQL query indexing → Study tracker.

- **Week 2: Backend Architecture & Networks** : Node.js/Express API design → Database design (NoSQL/MongoDB vs PostgreSQL) → Networks (HTTPS, SSL, DNS routing, firewalls).
- **Week 3: DevOps & Project** : Project Selection (e.g. Distributed Chat Portal, Cloud File Manager) → Docker containerization → AWS Deployment (EC2, S3, RDS) → CI/CD workflows.
- **Week 4: Information Security & Placement** : Web security (OWASP Top 10) → DSA (Search, Sort, Hash) → Aptitude → Mock Interviews → Portfolio.

2.2 Target Companies & Deliverables

- **Target Companies** : TCS, Infosys, Cognizant, Wipro, Accenture, AWS, Salesforce.
- **Deliverables** : Deployed AWS application, Security audit report, Resume, Portfolio.

3. Artificial Intelligence & Machine Learning (AI & ML)

Goal: Machine Learning Engineer, NLP Engineer, and computer vision roles.

3.1 Weekly Curriculum

- **Week 1: Math & Python Foundations** : Linear Algebra & Calculus → Probability & Statistics → Python for Data Science (NumPy, Pandas, Scipy) → Data cleaning.
- **Week 2: ML Model Implementation** : Supervised Learning (Regression, Decision Trees, SVM) → Unsupervised Learning (K-Means, PCA) → Evaluation metrics (Precision/Recall, ROC-AUC).
- **Week 3: Deep Learning & Projects** : Project Selection (e.g. Image Classification Model, LLM Conversational Bot) → Neural Networks (CNN, RNN, Transformers) → PyTorch/TensorFlow.
- **Week 4: MLOps & Placement** : Model Deployment (FastAPI, Streamlit) → Dockerizing ML pipelines → MLOps lifecycle → Aptitude → AI technical mocks.

3.2 Target Companies & Deliverables

- **Target Companies** : Microsoft, Google, NVIDIA, Fractal Analytics, Mu Sigma, Tech Startups.
- **Deliverables** : Deployed ML API model, Jupyter notebooks portfolio, Resume.

4. Data Science

Goal: Data Analyst, Data Engineer, and Business Intelligence Analyst.

4.1 Weekly Curriculum

- Week 1: Advanced SQL & Pandas : SQL Window functions, CTEs, optimization → Data Wrangling with Pandas → Exploratory Data Analysis (EDA).
- Week 2: Data Engineering & Warehouse : ETL/ELT pipeline design → Data Warehousing (Snowflake, Redshift) → NoSQL databases (MongoDB, Cassandra).
- Week 3: Big Data & Dashboards : Project Selection (e.g. Sales Prediction System, Live Web Scraping Dashboard) → Spark/Hadoop → BI Tools (Tableau/PowerBI).
- Week 4: Statistics & Placement : Statistical Hypothesis Testing (A/B testing) → ML basics → Aptitude → Data science mock interviews.

4.2 Target Companies & Deliverables

- Target Companies : EY, Deloitte, KPMG, PwC, Tiger Analytics, Fractal, Target.
- Deliverables : PowerBI/Tableau interactive dashboard, Spark ETL script, Resume.

5. Electronics & Communication Engineering (ECE)

Goal: Embedded Systems, VLSI Design, IoT, and hardware design.

5.1 Weekly Curriculum

- Week 1: Circuit Analysis & Analog : KVL/KCL, Network theorems → Semiconductor devices & diodes → BJT/MOSFET amplifiers → Digital design (Logic gates, FSMs).
- Week 2: Microcontrollers & Verilog : ARM Cortex/8051 architectures → Embedded C scripting → Verilog hardware description → DSP (Fourier & Z-transforms).
- Week 3: IoT & Hardware Project : Project Selection (e.g. Smart IoT Weather Station, FPGA Digital Clock) → Sensor interfacing (I2C, SPI, UART, LoRa) → LTspice circuit simulation.
- Week 4: VLSI Layout & Placement : CMOS VLSI layout rules → FPGA prototyping → Aptitude → Core ECE technical mock interviews.

5.2 Target Companies & Deliverables

- Target Companies : Intel, Qualcomm, Texas Instruments, NXP Semiconductors, Bosch, Samsung.
- Deliverables : Embedded firmware codebase, LTspice simulation layout, CAD schematic.

6. Electrical Engineering (EE)

Goal: Power Systems, Electrical Machines, Control Automation, and EV Powertrains.

6.1 Weekly Curriculum

- Week 1: Machines & Electromagnetics : Magnetic circuits & Transformers → DC Machines & AC Synchronous/Induction Motors → Transmission lines.
- Week 2: Power Electronics & Controls : Rectifiers, Inverters, Choppers → Power system fault analysis & protection → PID Control tuning (Bode plots, Nyquist).
- Week 3: MATLAB Simulink & EV Design : Project Selection (e.g. EV Motor Controller Simulink, Grid Solar Inverter) → MATLAB Power electronics model.
- Week 4: Smart Grid & Placement : Renewable integrations → Industrial Safety Electrical Codes (NEC) → Aptitude → Technical EE interview panel.

6.2 Target Companies & Deliverables

- Target Companies : Power Grid, NTPC, Tata Power, ABB, Schneider Electric, General Electric.
- Deliverables : MATLAB simulation model, Power schematics, CAD layouts.

7. Electrical & Electronics Engineering (EEE)

Goal: Industrial Automation (PLC/SCADA), Embedded systems, and Smart Grid technologies.

7.1 Weekly Curriculum

- Week 1: Core EEE Circuits : AC/DC Network analysis → Analog Op-Amps & Digital gates → Transformer & Induction Motor testing.
- Week 2: Control Systems & Power conversion : PID Controller structures → Power Electronic converters → Microcontroller interfacing (Arduino, ARM Cortex).
- Week 3: Industrial Automation Project : Project Selection (e.g. PLC-based Assembly Line Simulation, Smart Energy Meter) → PLC ladder logic → MATLAB modeling.
- Week 4: Industry 4.0 & Placement : Smart grid communication → SCADA dashboards → Aptitude & Reasoning → EEE Mock technical boards.

7.2 Target Companies & Deliverables

- Target Companies : Siemens, L&T Electrical, Alstom, Honeywell, Rockwell Automation.
- Deliverables : PLC Ladder Logic diagram, MATLAB simulation files, EEE Resume.

8. Mechanical Engineering

Goal: CAD Design, CAE Simulation, EV Powertrains, and Manufacturing.

8.1 Weekly Curriculum

- **Week 1: Core Mechanical Foundations** : Force Systems → Strength of Materials (Stress/Strain, Failure Theories) → Theory of Machines (TOM) (Gears, Cams) → Machine Design.
- **Week 2: Thermal & Manufacturing Systems** : Thermodynamics cycles → Heat & Mass Transfer → Fluid Mechanics → Manufacturing Processes (Casting, Machining).
- **Week 3: Design Tools & Project** : Project Selection (e.g. EV Go-Kart, Conveyor Design) → SolidWorks/CATIA Part & Assembly → GD&T → ANSYS CAE structural simulations.
- **Week 4: Future Tech & Placement** : EV Dynamics → Maintenance (TPM) → Quality (Six Sigma, SPC) → Technical Design interview grilling.

8.2 Target Companies & Deliverables

- **Target Companies** : Tata Motors, Mahindra, Maruti Suzuki, Bosch, Cummins, L&T, Siemens.
- **Deliverables** : Deployed CAD Assembly, CAE Simulation Report, Resume, GD&T sheet.

9. Civil Engineering

Goal: Structural analysis, Site Engineering, Estimation, and Quantity Surveying.

9.1 Weekly Curriculum

- **Week 1: Structural Foundations** : Strength of Materials (SFD, BMD) → Structural Analysis (Trusses) → Reinforced Concrete (RCC) limit states → Steel connections.
- **Week 2: Infrastructure Systems** : Geotechnical (Bearing capacity, Soil) → Surveying (Total Station, GPS) → Highway Pavements → Estimation & Costing (BOQ sheet).
- **Week 3: Software Tools & Project** : Project Selection (e.g. G+5 Building Design, Commercial Estimation) → AutoCAD plans → STAAD.Pro/ETABS loading → Primavera schedules.
- **Week 4: Site Safety & BIM** : BIM Revit overview → Construction Safety (OSHA) → NBC Codes → Technical Site Engineer Mock interview.

9.2 Target Companies & Deliverables

- **Target Companies** : L&T Construction, Shapoorji Pallonji, Tata Projects, AECOM, Jacobs.
- **Deliverables** : AutoCAD civil plans, ETABS structural model, BOQ estimation sheets.

10. Chemical Engineering

Goal: Process Design, Plant Operations, Refinery Operations, and Process Safety.

10.1 Weekly Curriculum

- **Week 1: Chemical Foundations** : Material & Energy balances → Chemical Thermodynamics → Fluid Flow & Pumps → Heat Exchangers → Mechanical operations.
- **Week 2: Process Engineering Core** : Mass Transfer (Distillation) → Reaction Engineering (CSTR/PFR design) → PID Control loops → PFD & P&ID interpretation.
- **Week 3: Aspen HYSYS Project** : Project Selection (e.g. Crude Distillation Simulation, Water Treatment plant) → Aspen HYSYS flowsheeting → HAZOP safety audits.
- **Week 4: Operations & Standards** : Refinery processes → Pharma GMP validation → OSHA/API standards → Technical Process engineering mock interview.

10.2 Target Companies & Deliverables

- **Target Companies** : Reliance Industries, IOCL, BPCL, BASF, Dow, Sun Pharma, Cipla.
- **Deliverables** : Aspen HYSYS flowsheet simulation file, HAZOP audit sheet, Process Report.

Comprehensive Application & Curriculum Workflows

Module 14: Fused Web Application, Course (Quest), & 10-Module Lifecycles

This document describes the foundational workflows governing the **PinIT Career OS (Version 2.0)** :

1. **Web Application Workflow** : Navigation states, routing structure, and screen transitions.
2. **Individual Course (Quest) Workflow** : The operational lifecycle of a student starting, compiling, and passing a single learning quest.
3. **The 10-Module Unified Workflow** : How all 10 platform architecture modules interact to take a student from registration to corporate placement.
4. **The 30-Day Branch-Specific Preparation Workflow** : Detailed chronological workflows for all 10 engineering departments (CSE, IT, AI & ML, Data Science, ECE, EE, EEE, Mechanical, Civil, Chemical).

1. The Web Application Workflow

PinIT's frontend is a Next.js single-page application (SPA) wrapped in a unified distraction-free shell ([AppShell.tsx](file:///c:/Users/vinay/Desktop/project/verify-pinit/firebase-deploy/src/components/ui/AppShell.tsx)).

```
```mermaid
```

```
graph TD
```

```
%% Public Routes
```

```
Landing["Landing Page (/)"] -->|Get Started| Signup["Signup (/signup)"]
```

```
Landing -->|Sign In| Login["Login (/login)"]
```

```
%% Onboarding Gateway
```

```
Signup --> Onboard["Onboarding Flow (/onboarding)"]
```

```
Login -->|If onboarding incomplete| Onboard
```

```
Login -->|If onboarding done| Dashboard["Command Center (/dashboard)"]
```

```
Onboard -->|STATE_3 Unlocked| Dashboard
```

```
%% Core OS Shell Navigation
```

```
Dashboard --> Quests["Quests Hub (/quests)"]
```

```
Dashboard --> Missions["Daily Missions (/missions)"]
```

```
Dashboard --> Vault["Evidence Vault (/vault)"]
```

```
Dashboard --> DNA["Career DNA (/career-dna)"]
```

```
Dashboard --> Opps["Opportunities (/opportunities)"]
```

```
Dashboard --> Profile["Profile & Mentor Config (/profile)"]
```

```
```
```

```
---
```

2. The Individual Course (Quest) Workflow

Every quest/course in PinIT follows a closed socratic learning loop. The browser sandbox handles compilation locally to reduce server cost to zero:

```
```mermaid
```

```
sequenceDiagram
```

```
autonumber
```

actor Student

participant Hub as Quests Hub (/quests)

participant Workspace as Editor Workspace (/quests/[id])

participant Sandbox as Local WASM Sandbox

participant AI as AI Socratic Mentor

participant DB as Supabase/Local Storage

Student->>Hub: Click "Start Quest"

Hub->>Workspace: Route to Quest Workspace (Deduct 5 Pins)

Workspace->>Student: Render objective checklist (Left) & seed code (Right)

Student->>Workspace: Write code or input design parameters

Student->>Workspace: Click "Run Tests"

Workspace->>Sandbox: Execute tests inside browser virtual environment

alt Tests Fail

Sandbox-->>Workspace: Return execution logs & assert errors

Workspace-->>Student: Display compilation errors & suggest hints

else Tests Pass

Sandbox-->>Workspace: Return 100% success signature

Workspace->>AI: Fetch socratic feedback tip

AI-->>Workspace: Return conceptual review tip

Workspace->>Student: Display success panel & AI feedback

Student->>Workspace: Click "Complete Quest"

Workspace->>DB: Write quest completion state, add +30 XP, award +10 Pins

DB-->>Hub: Unlock next sequential quest in timeline

end

```

3. The 10-Module Unified Workflow

This workflow represents the sequential interaction of all 10 platform architecture modules as a student prepares for corporate placement over 30 days:

...

[Module 1: Master State Machine]

|

v (Visitor logs in)

[Module 2: Onboarding Routes] → Express (Resume Parser) OR Deep (Scenario/Voice Diagnostic)

|

v (Calculates baseline traits)

[Module 4: Cognitive Diagnostics] & [Module 5: Mindset Archetypes & Scoring]

|

v (Dashboard HUD opens & priority tasks recommended)

[Module 6: Command Center HUD] → Toggles to WhatsApp-style Conversational Lite Mode if needed

|

v (Identifies skill gaps & compiles learning track)

[Module 7: sandboxed Workspace] → Local WebAssembly sandbox executes unit tests at \$0 cost

|

v (Deducts daily AI limits & awards operating Pins)

[Module 9: Fused Tokenomics] → Spend Pins to request custom topic missions

|

v (Uploads project files / credentials)

[Module 3: Trust-Verify Loop] → AI scanner checks signatures to raise Trust Score above 60%

|

v (Pre-renders voices & runs WebRTC assessments)

[Module 8: WebRTC Streams & S3 Caching] → Dynamic grilling by Coordinated AI panel

|

v (Dossier packet compiled: Code, DNA, credentials, voice)

[Module 10: Zero-Click Placement Delivery] → Pushes verified profile directly to recruiter ATS

...

4. The 30-Day Branch-Specific Preparation Workflow

Once onboarding completes, the platform launches a **30-Day Preparation Mode** tailored to the student's department. The student completes daily tasks in their sandbox, monitored by their **AI Mentor** and verified by **Evidence Vault** uploads.

```mermaid

graph TD

subgraph Phase 1: Foundation (Days 1-7)

A[Career Mapping & Resume Review] --> B[Daily Theory Quests]

B --> C[Weekly Foundation Test]

end

subgraph Phase 2: Core Mastery (Days 8-14)

C --> D[Advanced Core Concepts]

D --> E[Mid-Point Mock Interview]

end

subgraph Phase 3: Project Sizing (Days 15-21)

E --> F[Select Industry Project]

F --> G[CAD/Code Sandbox Simulations]

G --> H[Upload Proof to Vault]

end

subgraph Phase 4: Placement Launch (Days 22-30)

H --> I[Aptitude & Behavioral STAR]

I --> J[Coordinated AI Panel Assessment]

J --> K[Zero-Click Recruiter Push]

end

```

4.1 Computer Science and Engineering (CSE) Workflow

- **Days 1–7** : Map targets (SDE, Fullstack) → Resume rewrite → LinkedIn/GitHub optimization → Solve DSA Array/String quests → Execute SQL database queries.
- **Days 8–14** : Study OS process scheduling, DBMS transaction ACID controls, Computer Networks (HTTP/TCP), and SOLID design principles. Complete mock technical interviews.
- **Days 15–21** : Project Selection (SaaS, AI App, LMS) → System Design (caching, load balancing) → Implement Backend API and Auth → Configure Docker deployment → Commit to GitHub.
- **Days 22–30** : Practice aptitude tests → Record STAR behavioral interview responses → Solve DP/Graph challenges → Pass mock SDE panels → Finalize portfolio and activate recruiter push.

4.2 Information Technology (IT) Workflow

- **Days 1–7** : Audit career targets → Learn web basics (HTML5, CSS3, ES6 JavaScript) → Map DOM manipulations → Run SQL index optimization challenges.
- **Days 8–14** : Design RESTful backend APIs (Node.js/Express) → Structure NoSQL schemas (MongoDB) vs relational databases → Configure network firewalls, DNS routing, and HTTPS certificates.
- **Days 15–21** : Project Selection (Distributed Chat, Cloud File Explorer) → Setup AWS infrastructure (EC2, S3, RDS) → Write deployment Dockerfiles → Configure GitHub Action CI/CD pipelines.
- **Days 22–30** : Study OWASP Top 10 web vulnerabilities → Solve DSA search and sort challenges → Run aptitude tests → Complete IT infrastructure mocks → Finalize portfolio.

4.3 Artificial Intelligence & Machine Learning (AI & ML) Workflow

- **Days 1–7** : Review linear algebra, vector calculus, and statistics → Write Python data wrangling scripts (NumPy, Pandas) → Perform exploratory data feature scaling.
- **Days 8–14** : Implement supervised learning algorithms (Linear/Logistic regression, SVM, Random Forests) → Run unsupervised clustering (K-Means, PCA) → Measure precision-recall and ROC-AUC curves.
- **Days 15–21** : Project Selection (Image Classifier, LLM chatbot wrapper) → Configure Deep Learning layers (CNNs, Transformers) using PyTorch → Train models on local hardware mockups.
- **Days 22–30** : Deploy ML models as REST APIs using FastAPI → Dockerize model pipelines → Practice AI/ML technical interviews → Upload notebooks to GitHub and trigger push.

4.4 Data Science Workflow

- **Days 1–7** : Run advanced SQL queries (CTEs, Window functions, indexing) → Analyze datasets using Pandas → Generate exploratory data visualization charts.
- **Days 22–30** : Design ETL/ELT pipeline workflows → Connect streams to cloud warehouses (Snowflake, AWS Redshift) → Study NoSQL database storage (Cassandra, MongoDB).
- **Days 15–21** : Project Selection (Sales Prediction dashboard, web scraping pipeline) → Run big data processing scripts in Apache Spark → Design BI dashboards in Tableau/PowerBI.
- **Days 22–30** : Perform statistical hypothesis testing (A/B testing) → Complete business aptitude rounds → Run mock data interviews → Finalize portfolio reports.

4.5 Electronics & Communication Engineering (ECE) Workflow

- **Days 1–7** : Solve network theorems (KVL/KCL, Thevenin) → Analyze analog semiconductor circuits (BJTs, MOSFET amplifiers) → Design logic gates and finite state machines (FSMs).
- **Days 8–14** : Write Embedded C scripts for microcontrollers (ARM Cortex, 8051) → Describe digital circuits in Verilog/VHDL → Study DSP filters (Fourier, Z-transforms).
- **Days 15–21** : Project Selection (Smart IoT weather node, FPGA clock controller) → Connect sensors using protocols (I2C, SPI, UART) → Model schematic designs in LTspice.

- **Days 22–30** : Study CMOS VLSI layout rules → Prototype FPGA chips → Complete ECE core hardware mock interviews → Upload schematics to the Vault and trigger push.

4.6 Electrical Engineering (EE) Workflow

- **Days 1–7** : Map targets (Power grids, EV, control automation) → Analyze electromagnetic fields → Study DC/AC electrical machines and transformers.
- **Days 8–14** : Model power electronics components (choppers, rectifiers, inverters) → Run power system fault calculations → Tune PID controller parameters (Bode plots, Nyquist).
- **Days 15–21** : Project Selection (EV motor control simulator, grid solar inverter simulation) → Build circuits inside MATLAB Simulink → Settle hardware component ratings.
- **Days 22–30** : Study smart grid communication and renewable grid integrations → Audit electrical safety standards (NEC codes) → Pass EE mock panels → Upload MATLAB designs.

4.7 Electrical & Electronics Engineering (EEE) Workflow

- **Days 1–7** : Solve AC/DC network circuit equations → Design analog amplifiers (Op-Amps) and digital logic controllers → Conduct induction motor parameter tests.
- **Days 8–14** : Wire power converters (buck-boost controllers) → Design closed-loop control systems → Write microcontroller code (Arduino, ARM Cortex).
- **Days 15–21** : Project Selection (PLC industrial assembly simulator, smart power meter) → Program PLC ladder logic → Simulate circuits in MATLAB.
- **Days 22–30** : Integrate smart SCADA communication dashboards → Study IoT controller nodes → Complete EEE technical mock interviews → Upload PLC logic to the Vault.

4.8 Mechanical Engineering Workflow

- **Days 1–7** : Audit targets (Design, Automotive, Manufacturing) → Solve Strength of Materials calculations (failure theories) → Model TOM mechanism motions (gears, cams).
- **Days 8–14** : Study thermodynamic cycles and heat transfer (conduction, convection) → Model fluid mechanics pipe flows → Analyze manufacturing processes (casting, welding).
- **Days 15–21** : Project Selection (EV Go-Kart, HVAC system, Conveyor design) → Model parts in SolidWorks/CATIA → Apply GD&T drawings → Run structural/thermal CAE in ANSYS.
- **Days 22–30** : Study EV drivetrain dynamics and smart robotics → Implement Lean Manufacturing (Six Sigma, TPM) → Upload CAD models to the Vault → Pass mock design interviews.

4.9 Civil Engineering Workflow

- **Days 1–7** : Audit targets (Structures, Site, Estimation) → Solve strength of materials beam diagrams (SFD, BMD) → Analyze trusses → Design RCC concrete limit states and steel welds.
- **Days 8–14** : Calculate soil bearing capacities → Survey total station and GPS coordinates → Design highway pavement pavements → Compile estimation takeoffs.

- **Days 15–21** : Project Selection (G+5 residential building, commercial BOQ) → Draft blueprints in AutoCAD Civil → Apply loads in STAAD.Pro → Analyze seismic safety in ETABS.
- **Days 22–30** : Manage scheduling in Primavera/MS Project → Run Revit BIM inspections → Audit IS 456/IS 800 code standards → Pass technical site engineer mock interviews.

4.10 Chemical Engineering Workflow

- **Days 1–7** : Calculate process material and energy balances → Solve phase equilibria thermodynamics → Model fluid pipe flows and pumps → Study heat exchangers.
- **Days 8–14** : Analyze distillation column mass transfer → Model CSTR/PFR reactors → Design PID controller loops → Interpret process flow diagrams (PFD & P&ID).
- **Days 15–21** : Project Selection (Crude oil distillation simulation, water treatment design) → Simulate process flowsheets in Aspen HYSYS → Size equipment capacities → Audit HAZOP safety rules.
- **Days 22–30** : Study refinery petrochemical processes, pharma GMP guidelines, and green hydrogen/biofuel sustainability. Upload HYSYS files, pass HAZOP technical mocks, and trigger push.